A Cardiac Electrophysiology Forward Pipeline and Single-Level Schwarz Preconditioning

Continuous & discrete formulation · the overlap anomaly · cross-system preconditioning

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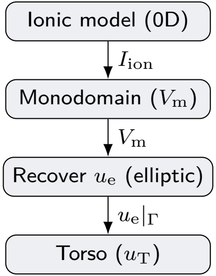
Part I–II: ionic model $\to$ monodomain $\to$ extracellular recovery $\to$ torso (real geometry).

Part III: overlap-dependent CG iteration growth in ASM + ICC (abstract test problems).

Part IV: can one system precondition another? — theory, prior art, and transferable ideas.

Code & data: `github.com/tianhaomahpc-ops/cg-asm-icc-research`

Part I — Continuous problem: domains and notation



$V_{\text{m}}, u_{\text{e}}$	transmembrane / extracellular potential (in Ω_{H})
u_{T}	torso potential (in Ω_{T})
$\boldsymbol{w}, \boldsymbol{z}$	gating variables / ion concentrations
$I_{\text{ion}}, I_{\text{stim}}$	total ionic current / stimulus current
χ, C_{m}	surface-to-volume ratio, membrane capac.
$\boldsymbol{\sigma}_{\text{i}}, \boldsymbol{\sigma}_{\text{e}}, \boldsymbol{\sigma}_{\text{m}}$	intra / extra / monodomain conductivities
σ_{T}	torso conductivity (uniform scalar)
Γ, $\partial\Omega_{\text{T}}^{\text{ext}}$	heart–torso interface, body surface
$\mathbf{M}, \mathbf{K}_{\boldsymbol{\sigma}}$	P1 mass / stiffness ($\boldsymbol{\sigma}$-weighted)
Δt, n	time step (0.01 ms), time index

Heart Ω_{H} and torso Ω_{T} are *two meshes sharing* the heart surface $\Gamma = \partial\Omega_{\text{H}}$.

Parameters: Niederer et al. (2011) benchmark; cell model: ten Tusscher–Panfilov (2006, "TP06").

1. Ionic model (TP06, 0D cell kinetics)

state split into gating variables w and concentrations z (the only ODEs written here)

At each spatial point, the membrane current is governed by gating and concentration ODEs:

$$\textbf{gating:}\ \ \dot{w}_j = \frac{w_{j,\infty}(V_\mathrm{m}) - w_j}{\tau_{w_j}(V_\mathrm{m})}, \quad \textbf{concentrations:}\ \ \dot{z}_k = f_{z_k}(V_\mathrm{m}, \boldsymbol{w}, \boldsymbol{z}),$$

with the total ionic current assembled from the TP06 channel/pump/exchanger currents

$$I_\mathrm{ion} = I_\mathrm{ion}(V_\mathrm{m}, \boldsymbol{w}, \boldsymbol{z}).$$

- $\boldsymbol{w} =$ voltage-gated gating variables (Hodgkin–Huxley form, hence the w_∞, τ_w structure).
- $\boldsymbol{z} =$ intracellular ionic concentrations (and related state).
- In tissue, V_m is *not* a 0D ODE: it is advanced by the monodomain PDE (next slide); I_ion enters that PDE as the reaction term.
- Sign convention and rate functions: as in the TP06 model with Niederer-benchmark parameters.

2. Monodomain equation (reaction–diffusion for V_{m})

in the heart Ω_{H}, with an electrically insulated surface

$$\boxed{\chi\left(C_{\mathrm{m}}\,\frac{\partial V_{\mathrm{m}}}{\partial t} + I_{\mathrm{ion}}(V_{\mathrm{m}},\boldsymbol{w},\boldsymbol{z})\right) = \nabla\cdot(\boldsymbol{\sigma}_{\mathrm{m}}\nabla V_{\mathrm{m}}) + I_{\mathrm{stim}}} \qquad \text{in } \Omega_{\mathrm{H}}\times(0,T]$$

boundary condition: $(\boldsymbol{\sigma}_{\mathrm{m}}\nabla V_{\mathrm{m}})\cdot\mathbf{n} = 0$ on $\partial\Omega_{\mathrm{H}} = \Gamma$ (no extracellular coupling).

- $\boldsymbol{\sigma}_{\mathrm{m}} = \boldsymbol{\sigma}_{\mathrm{i}}(\boldsymbol{\sigma}_{\mathrm{i}}+\boldsymbol{\sigma}_{\mathrm{e}})^{-1}\boldsymbol{\sigma}_{\mathrm{e}}$: monodomain conductivity (harmonic combination of intra/extra).
- $\chi, C_{\mathrm{m}}, \boldsymbol{\sigma}_{\mathrm{m}}$: Niederer et al. (2011) benchmark values (anisotropic, fiber-aligned $\boldsymbol{\sigma}_{\mathrm{m}}$).
- I_{stim} : applied stimulus (location / waveform omitted here by request).
- Coupled to the cell model through I_{ion}; this is the parabolic stage of the pipeline.

3. Recover the extracellular potential u_e

elliptic problem in the heart Ω_H — pure Neumann, singular

$$\boxed{-\nabla\cdot\big((\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e})\nabla u_\mathrm{e}\big) = \nabla\cdot(\boldsymbol{\sigma}_\mathrm{i}\nabla V_\mathrm{m})} \qquad \text{in } \Omega_\mathrm{H}$$

boundary condition (pure Neumann): $\big((\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e})\nabla u_\mathrm{e}\big)\cdot\mathbf{n} = -(\boldsymbol{\sigma}_\mathrm{i}\nabla V_\mathrm{m})\cdot\mathbf{n}$ on Γ.

- Operator $-\nabla\cdot((\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e})\nabla\,\cdot\,)$ with all-Neumann data is **singular**: $\ker = \{\text{constants}\}$.
- Solvability: the right-hand side must have zero mean over Ω_H (current balance); the solution is fixed by the **mean-zero gauge** $\int_{\Omega_\mathrm{H}} u_\mathrm{e} = 0$.
- This is the elliptic “bidomain-recovery” stage; V_m from stage 2 is its data.
- In the linear-algebra study (Part III) this is the prototype of the *singular all-Neumann* system.

4. Torso model (passive volume conductor)

elliptic problem in the torso Ω_{T} — one-way coupling from the heart

$$\boxed{-\nabla\cdot(\sigma_{\mathrm{T}}\nabla u_{\mathrm{T}}) = 0} \qquad \text{in } \Omega_{\mathrm{T}}, \qquad \sigma_{\mathrm{T}} = \text{const (uniform)}$$

coupling (Dirichlet, one-way): $u_{\mathrm{T}} = u_{\mathrm{e}}$ on Γ; **body surface:** $(\sigma_{\mathrm{T}}\nabla u_{\mathrm{T}})\cdot\mathbf{n} = 0$ on $\partial\Omega_{\mathrm{T}}^{\mathrm{ext}}$.

- One-way: the heart supplies $u_{\mathrm{e}}|_{\Gamma}$ as **Dirichlet** data; no feedback to the heart.
- Mixed BC (Dirichlet on Γ, Neumann on the body surface) $\Rightarrow$ **non-singular** elliptic solve.
- Ω_{H} and Ω_{T} are two meshes that share the surface Γ; $u_{\mathrm{e}}|_{\Gamma}$ is transferred across the conforming interface.
- Prototype (Part III) of the *mixed Dirichlet/Neumann* system (well-posed, ill-conditioned).

Part II — Discretization: common setup

- **Space:** P1 (piecewise-linear) finite elements on tetrahedral meshes $\mathcal{T}_h$ of Ω_{H} and Ω_{T}; nodal unknowns.
- **Matrices** (per domain / conductivity):

$$(\mathbf{M})_{ab} = \int \varphi_a \varphi_b, \qquad (\mathbf{K}_{\boldsymbol{\sigma}})_{ab} = \int \boldsymbol{\sigma} \nabla \varphi_a \cdot \nabla \varphi_b \quad \text{(SPD, } \mathbf{K}_{\boldsymbol{\sigma}} \leftrightarrow -\nabla \cdot (\boldsymbol{\sigma} \nabla \,\cdot)\text{)}.$$

- **Time:** $\Delta t = 0.01\,\mathrm{ms}$; superscript n denotes time level $t_n = n\Delta t$.
- **Splitting (IMEX):** the diffusion is treated implicitly (Crank–Nicolson), the reaction I_{ion} as an **explicit source**; the gating/concentration ODEs are integrated pointwise.
- **Initial conditions:** $V_{\mathrm{m}}^0 = -83\,\mathrm{mV}$ (resting), $\boldsymbol{w}^0, \boldsymbol{z}^0 =$ TP06 resting state, $u_{\mathrm{e}}^0 = 0$, $u_{\mathrm{T}}^0 = 0$.

$1'$. Ionic time integration: Rush–Larsen + forward Euler

pointwise at every node, advancing $t_n \to t_{n+1}$ from V_m^n

Gating w — Rush–Larsen (exact for the frozen-V_m linear ODE; unconditionally stable):

$$w_j^{n+1} = w_{j,\infty}(V_\mathrm{m}^n) + \big(w_j^n - w_{j,\infty}(V_\mathrm{m}^n)\big)\,\exp\Big(-\frac{\Delta t}{\tau_{w_j}(V_\mathrm{m}^n)}\Big).$$

Concentrations z — forward Euler:

$$z_k^{n+1} = z_k^n + \Delta t\, f_{z_k}(V_\mathrm{m}^n, \boldsymbol{w}^n, \boldsymbol{z}^n).$$

Explicit ionic current (the source fed to the monodomain solve):

$$I_\mathrm{ion}^n := I_\mathrm{ion}(V_\mathrm{m}^n, \boldsymbol{w}^{n+1}, \boldsymbol{z}^{n+1}).$$

Rush–Larsen handles the stiff gating kinetics; forward Euler suffices for the slower concentrations.

2′. Monodomain: Crank–Nicolson diffusion, explicit reaction $\Rightarrow$ Sys 1

P1 FEM in space; the reaction I_{ion}^{n} is a known right-hand side

$$\boxed{\underbrace{\left(\frac{\chi C_{\mathrm{m}}}{\Delta t}\mathbf{M} + \frac{1}{2}\mathbf{K}_{\boldsymbol{\sigma}_{\mathrm{m}}}\right)}_{\mathbf{A}_1\ \text{(SPD)}}\mathbf{V}^{n+1} = \left(\frac{\chi C_{\mathrm{m}}}{\Delta t}\mathbf{M} - \frac{1}{2}\mathbf{K}_{\boldsymbol{\sigma}_{\mathrm{m}}}\right)\mathbf{V}^{n} - \chi\,\mathbf{M}\,I_{\text{ion}}^{n} + \mathbf{M}\,I_{\text{stim}}^{n}}$$

- **Sys 1** $= \mathbf{A}_1 = \frac{\chi C_{\mathrm{m}}}{\Delta t}\mathbf{M} + \frac{1}{2}\mathbf{K}_{\boldsymbol{\sigma}_{\mathrm{m}}}$ — the matrix our study calls $\frac{1}{\Delta t}M + \frac{1}{2}K$.
- At $\Delta t = 0.01\,\text{ms}$ the **mass term dominates** $\Rightarrow$ well-conditioned $\Rightarrow$ very few CG iterations.
- Explicit reaction keeps the left-hand side *linear, SPD, and constant in time* $\Rightarrow$ the preconditioner is built once and reused every step.

3′. Recover u_e: P1 FEM $\Rightarrow$ Sys 2 (singular)

$$\boxed{\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e}}\,\mathbf{u}_\mathrm{e}^{n+1} = -\,\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}}\,\mathbf{V}^{n+1}} \qquad \text{then re-gauge} \quad \mathbf{u}_\mathrm{e}^{n+1} \leftarrow \mathbf{u}_\mathrm{e}^{n+1} - \mathrm{mean}(\mathbf{u}_\mathrm{e}^{n+1}).$$

- **Sys 2** $= \mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e}}$: symmetric positive *semi*-definite, $\ker = \mathrm{span}\{\mathbf{1}\}$.
- Right-hand side $-\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}}\mathbf{V}^{n+1}$ is consistent (in range); the constant null mode is removed from the RHS and from the iterate (mean-zero gauge).
- Solved by preconditioned CG on the consistent singular system (no operator shift).

4′. Torso: P1 FEM with interface Dirichlet data $\Rightarrow$ Sys 3-type

$$\boxed{\mathbf{K}_{\sigma_{\mathrm{T}}}\,\mathbf{u}_{\mathrm{T}}^{n+1} = \mathbf{0}} \qquad \text{subject to} \quad \mathbf{u}_{\mathrm{T}}^{n+1}\big|_{\Gamma} = \Pi\big(\mathbf{u}_{\mathrm{e}}^{n+1}\big|_{\Gamma}\big),$$

with $(\sigma_{\mathrm{T}}\nabla u_{\mathrm{T}})\cdot\mathbf{n} = 0$ on the body surface $\partial\Omega_{\mathrm{T}}^{\mathrm{ext}}$.

- Π : transfer of u_{e} from the heart-surface trace to the torso mesh (shared Γ, conforming).
- Dirichlet rows on Γ make the system **non-singular**; reduced operator is SPD.
- Mixed Dirichlet (Γ) / Neumann (body) $\Rightarrow$ the well-posed but ill-conditioned elliptic prototype studied abstractly in Part III.

One time step: the full update $t_n \to t_{n+1}$

Given $V_\mathrm{m}^n, \boldsymbol{w}^n, \boldsymbol{z}^n$ (and $u_\mathrm{e}^n, u_\mathrm{T}^n$):

(1) **Gating** (Rush–Larsen): $\boldsymbol{w}^{n+1} \leftarrow$ exact exponential update from V_m^n.

(2) **Concentrations** (forward Euler): $\boldsymbol{z}^{n+1} \leftarrow \boldsymbol{z}^n + \Delta t\, f_{\boldsymbol{z}}(V_\mathrm{m}^n, \boldsymbol{w}^n, \boldsymbol{z}^n)$.

(3) **Ionic source** (explicit): $I_\mathrm{ion}^n \leftarrow I_\mathrm{ion}(V_\mathrm{m}^n, \boldsymbol{w}^{n+1}, \boldsymbol{z}^{n+1})$.

(4) **Monodomain** (Sys 1): solve $\mathbf{A}_1 \mathbf{V}^{n+1} = \left(\frac{\chi C_\mathrm{m}}{\Delta t}\mathbf{M} - \frac{1}{2}\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{m}}\right)\mathbf{V}^n - \chi \mathbf{M} I_\mathrm{ion}^n + \mathbf{M} I_\mathrm{stim}^n$.

(5) **Recover** u_e (Sys 2): solve $\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}+\boldsymbol{\sigma}_\mathrm{e}} \mathbf{u}_\mathrm{e}^{n+1} = -\mathbf{K}_{\boldsymbol{\sigma}_\mathrm{i}} \mathbf{V}^{n+1}$; re-gauge to mean zero.

(6) **Torso** (Sys 3): solve $\mathbf{K}_{\sigma_\mathrm{T}} \mathbf{u}_\mathrm{T}^{n+1} = \mathbf{0}$ with $\mathbf{u}_\mathrm{T}|_\Gamma = \Pi(\mathbf{u}_\mathrm{e}^{n+1}|_\Gamma)$.

All four solves are performed every step. The three linear systems (Sys 1/2/3) share the heart mesh and sparsity; their matrices are constant in time, so each preconditioner is set up once and reused.

Initial conditions: $V_\mathrm{m}^0 = -83\,\mathrm{mV}$, $\boldsymbol{w}^0, \boldsymbol{z}^0 =$ TP06 resting state, $u_\mathrm{e}^0 = u_\mathrm{T}^0 = 0$.

Part III — The overlap anomaly (abstract test problems)

not the real geometry: clean prototypes that isolate the linear algebra

Solver under study: CG + additive Schwarz (`PCASM`, BASIC variant) + subdomain ICC(L), overlap O.

Prototype problems on the unit cube $[0,1]^3$, P1 tetrahedra, $n_x{=}48$ ($117{,}649$ DOF), 4 MPI ranks, METIS:

- mixed Dirichlet/Neumann Laplace (1 Dirichlet + 5 Neumann faces) — analogue of the torso solve;
- all-Neumann singular Laplace ($\ker = \{\text{const}\}$) — analogue of the u_e recovery;
- reaction–diffusion $\frac{1}{\Delta t}\mathbf{M} + \frac{1}{2}\mathbf{K}$ — analogue of the monodomain step.

Expectation (classical overlapping Schwarz, exact local solves): more overlap $\Rightarrow$ smaller condition number $\Rightarrow$ *fewer* iterations.

The suspicious result: more overlap $\Rightarrow$ *more* iterations

BASIC ASM + CG, mixed-BC Laplace; CG iteration count

$O\backslash L$	0	1	2	3	4	5	6
0	132	102	87	77	72	66	64
1	148	102	92	75	65	56	50
2	179	117	90	74	65	57	51
3	199	130	103	80	59	56	50
trend in O	$\uparrow$	$\uparrow$	$\uparrow$	$\approx$	$\downarrow$	$\downarrow$	$\downarrow$

- At low fill ($L \leq 3$): increasing overlap **raises** the iteration count ($132 \to 199$ at $L=0$) — opposite to theory.
- The professor's objection: "more overlap should help." It does — but only once L is large enough.

Conjecture: two coupled causes

the abstract Schwarz condition-number bound

$$\kappa(\mathbf{B}^{-1}\mathbf{A}) \;\le\; \underbrace{C_0^2}_{\substack{\text{stable splitting}\\ \delta\uparrow\Rightarrow\downarrow}} \cdot \underbrace{\omega}_{\substack{\text{(A) local solve}\\ \text{exact}=1,\ \text{ICC}>1}} \cdot \underbrace{N_c}_{\substack{\text{(B) overlap colouring}\\ \delta\uparrow\Rightarrow\uparrow}} \qquad \text{(Toselli \& Widlund, Thm. 2.7)}$$

- **(A) inexact subdomain solve.** ICC is not the exact block inverse: $\omega > 1$, and it worsens as the overlapped block grows.
- **(B) overlap duplicate counting.** BASIC over-counts the overlap region: $\sum_i R_i^\top R_i = \mathrm{diag}(m_k)$, and the colouring number N_c grows with overlap.
- Larger overlap shrinks C_0^2 but *inflates* A $\times$ B; at low ICC the latter wins $\Rightarrow$ net increase.

Pinning it down: remove the inexactness, the trend reverses

same problem, only the subdomain solver changes; CG iterations

subdomain solver	$O{=}0$	$O{=}1$	$O{=}2$	trend
exact Cholesky	52	34	30	$\downarrow$
ICC(0)	132	148	179	$\uparrow$

- Exact local solve ($\omega{=}1$): overlap helps, $52 \to 30$ — classical behaviour restored.
- Raising ICC fill drives $\omega \to 1$: the overlap trend **flips at** $L \approx 4$–5 (see prior table).
- Confirms factor (A): the anomaly needs an *inexact* subdomain solve.

The same anomaly appears on all three abstract prototypes (mixed-BC, singular, reaction–diffusion).

The fix: scaled ASM removes the duplicate counting (factor B)

$\mathbf{B}_{\mathrm{sASM}}^{-1} = \mathbf{D}^{-1/2}\mathbf{B}_{\mathrm{BASIC}}^{-1}\mathbf{D}^{-1/2}$, $\mathbf{D} = \mathrm{diag}(m_k)$ **(preconditioner** B**, not the mass matrix) — symmetric, CG-valid**

O	0	1	2
BASIC	132	148	179
sASM	132	**104**	**103**

- The m_k-fold over-count of the overlap is cancelled by $1/\sqrt{m_k}$ on each side.
- $\mathbf{D}$ diagonal $\Rightarrow$ operator stays symmetric $\Rightarrow$ CG remains valid; $O{=}0 \Rightarrow \mathbf{D}{=}\mathbf{I}$ (=BASIC).
- Overlap becomes useful again: $132 \rightarrow 104 \rightarrow 103$.

Verified across all three prototypes and bit-identically on two independent implementations (MFEM and pure PETSc). Stronger single-level variants (Chebyshev block solve; graded-weight & symmetrized-multiplicative RAS) push iterations lower still.

Part IV — Can one system precondition another?

small Δt: Sys 1 is cheap; the cost is in the elliptic Sys 2 / Sys 3

Naive idea: use Sys 1 ($\frac{\chi C_\mathrm{m}}{\Delta t}\mathbf{M} + \frac{1}{2}\mathbf{K}$) to precondition the elliptic $\mathbf{K}$. **It fails spectrally:**

$$\text{generalized eigenvalues} \quad \lambda \longmapsto \frac{\lambda}{1/\Delta t + \lambda/2} : \qquad \lambda_{\min} \text{ collapses by a factor } \approx \Delta t.$$

Equivalent-operator theory says exactly why (the precise criterion for "B preconditions A"):

- **same principal part** (Faber–Manteuffel–Parter 1990): $\frac{1}{\Delta t}\mathbf{M} + \frac{1}{2}\mathbf{K}$ and $\mathbf{K}$ are *not* equivalent as $\Delta t \to 0$.
- **same boundary conditions** (Manteuffel–Parter 1990): an all-Neumann operator cannot precondition a mixed-BC one.
- **parameter-robust norms** (Mardal–Winther 2011): in $L^2 \cap \sqrt{\Delta t}\, H^1$, Sys 1 is uniformly easy — and its inverse carries no H^1 (low-frequency) information.

What others do: precondition by a designed *nearby* operator

- **Cardiac precedent — Gerardo-Giorda et al. (JCP 2009):** the *monodomain* operator preconditions the *bidomain* system. It works because it **keeps the elliptic principal part** and only simplifies the coefficients — the legal version of "front system $\to$ back system."
- **Cardiac practice (CARP / openCARP; Mardal–Nielsen–Cai–Tveito 2007):** the elliptic block is the bottleneck and is preconditioned by a multigrid *of its own operator*; the parabolic step is trivial (Jacobi / lumped mass, $\kappa \leq d+2$). Setup amortized across time steps.
- **Shifted-Laplacian lesson (Erlangga–Vuik–Oosterlee; Gander–Graham–Spence 2015):** a nearby-operator preconditioner only works if the shift stays spectrally close. $1/\Delta t$ is the "too-large shift" — easy to invert, far from $\mathbf{K}$.

Lesson: transfer the *principal part* (the stiffness $\mathbf{K}$ / a Laplacian), not the mass-dominated Sys 1.

Transferable idea 1: slow subspaces, not operators (deflation)

what survives a cross-operator jump is the low-frequency end

- The whole family $\alpha\mathbf{M} + \beta\mathbf{K}$ shares **one set of generalized eigenvectors** $(\mathbf{K}, \mathbf{M})$; only the eigenvalues change. So Sys 1 and Sys 2 (same mesh, same BC) have the **same eigenvectors** — the operator inverse does not transfer, but the *subspace* does, exactly.
- **Deflated CG** (Saad–Yeung–Erhel–Guyomarc'h 2000): extract approximate *smallest*-eigenvalue vectors while solving the cheap system, deflate them from the hard one. The small (smooth) modes are precisely the ones the mass term / Δt-scaling leaves nearly invariant.
- **Singular Sys 2 for free:** the constant null vector *is* the slowest mode; using it (or subdomain-constants) as a deflation / coarse vector regularizes the singularity *and* deflates the slowest mode (Tang–Vuik deflated ICCG).
- Cross-BC (Sys 2 $\rightarrow$ Sys 3) is approximate; the perturbation bounds of Gaul–Gutknecht–Liesen–Nabben (2013) quantify how the gain decays.

Transferable idea 2: reuse the solution history

cheap / earlier data accelerates the harder, related solve

- **Fischer projection initial guess (1998; PETSc `KSPGUESSFISCHER`):** keep an orthonormal basis of recent time-step solutions, project the current RHS onto it for a near-optimal start. Needs only a *fixed* operator with successive RHS — exactly Sys 2 / Sys 3 across time. Nearly free; $\sim 2\times$ reported. *First thing to try.*
- **POD-as-deflation (Astudillo–Díaz-Cortés–Vuik):** build deflation vectors from a *moving window* of recent solution snapshots; deflate the hard elliptic solves. Iterations cut to $\sim$10–30% of the baseline (a handful in favourable cases), up to $\sim 5\times$ speed-up. **CG stays the solver** — POD only supplies a subspace (compatible with "no POD as the solver").
- These two compose: a better *start* (Fischer) + a better *spectrum* (deflation).

Transferable idea 3: share structure; build the singular hierarchy on a shift

Structure sharing (unconditional win).

- Sys 1/2/3 share mesh & sparsity (heart mesh) $\Rightarrow$ **symbolic factorization, METIS partition, overlap IS, multiplicity / graded weights** built *once*, reused.
- Only the numeric factorization differs per operator (Benzi–Bertaccini shifted-factorization updates).

Singular Sys 2: shift only to build.

- Kuchta–Mardal–Mortensen (2019): build the AMG/ICC hierarchy on a *nonsingular* shift $\mathbf{K} + \sigma\,\mathbf{I}$, then solve the *original* singular system.
- Identity shift $\mathbf{K} + \sigma\mathbf{I}$ **beats** mass shift $\mathbf{K} + \sigma\mathbf{M}$ (preserves the constant kernel); take $\sigma \sim \lambda_1$, *not* $1/\Delta t$.

Caveat: in the real geometry the torso is a *separate mesh* from the heart — structure sharing applies to Sys 1$\leftrightarrow$Sys 2 (heart); the torso shares only the methodology and the Fischer initial guess.

Summary and a concrete roadmap

Verdict. The operator inverse does *not* transfer (same-principal-part & same-BC are theorems; $1/\Delta t$ is too large a shift). Three things *do* transfer:

what transfers	mechanism	for our systems
slow subspace	deflated CG; constant null vector	Sys 2 (free), Sys 1$\rightarrow$Sys 2 (exact eigvecs)
solution history	Fischer guess; POD-as-deflation	Sys 2, Sys 3 across time steps
structure	symbolic factor. / partition / weights	Sys 1$\leftrightarrow$Sys 2 (shared heart mesh)

Three experiments, runnable in our harness:

1. Turn on `KSPGUESSFISCHER` for Sys 2 / Sys 3 across time steps — measure the speed-up.
2. Sys 2: constant-vector deflation + hierarchy built on $\mathbf{K} + \sigma\mathbf{I}$ — compare with the mean-zero baseline.
3. Prototype "Sys 1 low modes $\rightarrow$ Sys 2 deflation" to confirm the exact-eigenvector transfer.